

DTM Drainage AI

Automated Terrain Modelling · Waterlogging Risk · Drainage Network Design

End-to-end pipeline from airborne LiDAR point clouds — SVAMITVA scheme

MoPR Geospatial Intelligence Hackathon · Problem Statement 2

Team: charansaiponnada · IIT Tirupati

4 villages processed across Gujarat & Punjab · Devdi Khapreta Dhal Hoshiarpur Chakhirasingh

Raw LiDAR → DTM + Risk Map + Costed Drainage in one automated run

The Problem

Rural India needs scalable, data-driven flood-mitigation planning

- MoPR's SVAMITVA scheme captures drone LiDAR over thousands of village abadi areas — a vast, underused 3-D terrain asset.
- Manual drainage planning is slow, inconsistent, and cannot scale to 600,000+ villages.
- Monsoon waterlogging damages property and health; planners lack objective, repeatable hotspot maps.
- Raw point clouds are not directly usable — they must be turned into terrain models, risk maps and engineered designs.

#	Deliverable required by PS-2
1	High-resolution Digital Terrain Model (≤ 0.5 m) from raw LiDAR
2	AI/ML waterlogging risk map over terrain-derived features
3	Cost-optimised drainage network with hydraulic design
4	All outputs in open OGC formats (COG raster + GeoPackage vector)

Our Solution — One Automated Pipeline

Raw LAS/LAZ in · OGC-compliant DTM, risk map and costed drainage out

Fully automated

- Single command runs all 6 stages
- No manual GIS steps
- Batch mode for many villages

Physically grounded

- SMRF ground filter + IDW DTM
- D8 hydrology, TWI, depressions
- Manning's hydraulics + costing

Scalable & open

- Auto-tiling for any file size
- Cloud-Optimized GeoTIFF + GPKG
- Transfers across states

4

villages, 2 states

314M

LiDAR points processed

135 km

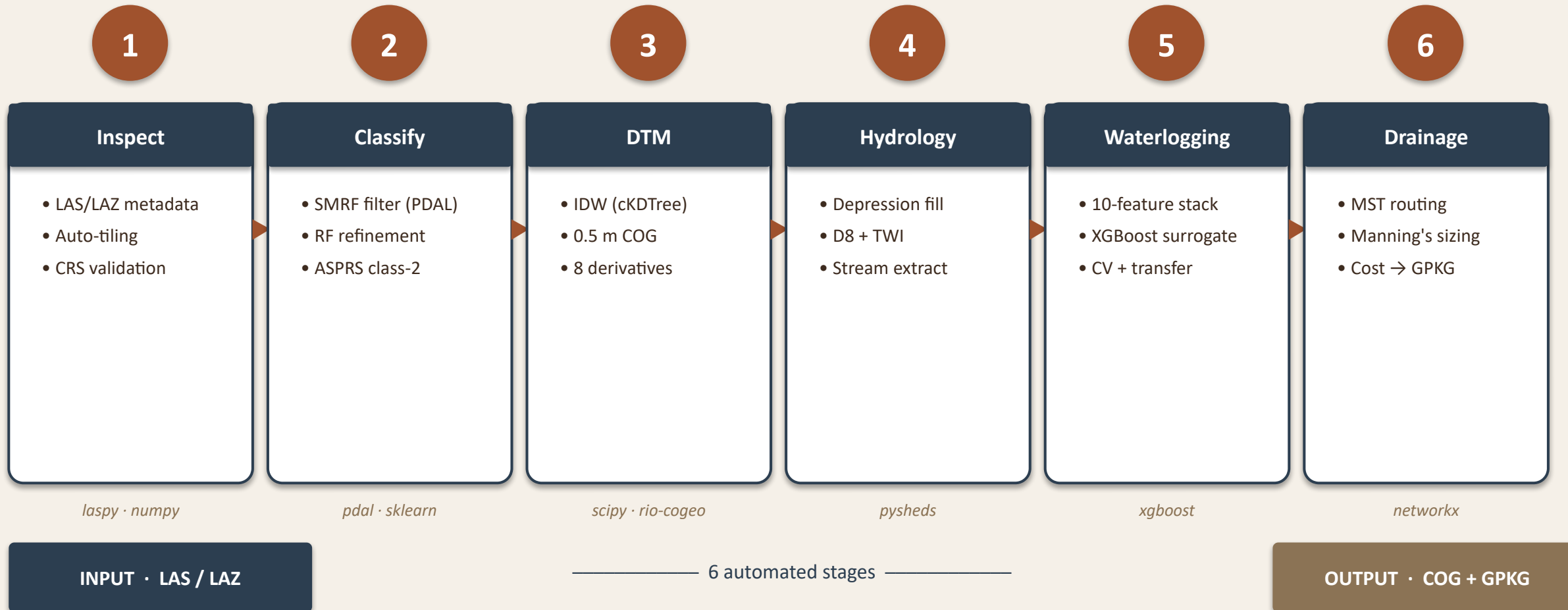
drainage designed

₹1097L

costed network

System Architecture — 6-Stage Pipeline

One command · raw LiDAR in → OGC-compliant DTM, risk map and costed drainage out



Methodology — Terrain Modelling (Stages 1–3)

1 · Inspect

- Read LAS/LAZ headers (laspy)
- Validate CRS, density, bounds
- Auto-tile big files (200 m + 25 m buffer)

2 · Classify

- SMRF morphological filter (PDAL)
- Random-Forest refinement
- Output ASPRS class-2 ground LAS

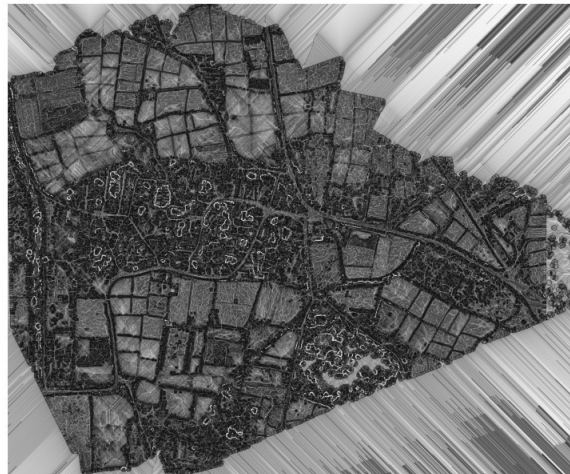
3 · DTM

- IDW interpolation (cKDTree, k=16)
- 0.5 m Cloud-Optimized GeoTIFF
- 8 terrain derivatives computed

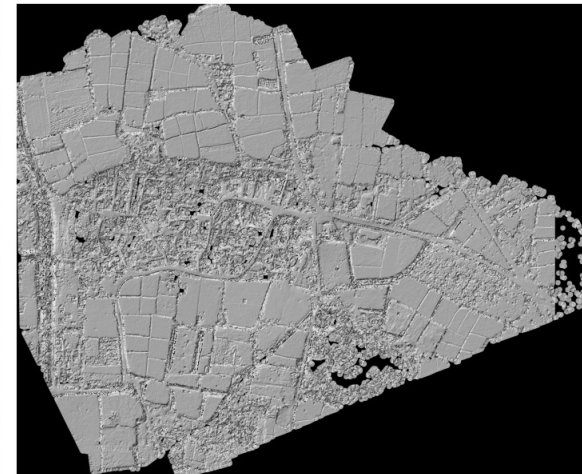
Slope



Topographic Wetness Index



Hillshade



Terrain derivatives from the DTM: slope · Topographic Wetness Index · hillshade

Methodology — Hydrology, AI & Drainage (Stages 4–6)

4 · Hydrology

- Depression filling (pysheds)
- D8 flow direction + accumulation
- $TWI = \ln(\alpha / \tan \beta)$
- Stream network extraction

5 · Waterlogging

- 10-feature terrain stack
- XGBoost risk surrogate
- Physically-grounded risk index
- 5-fold CV + cross-village test

6 · Drainage

- Minimum Spanning Tree routing
- Rational method: $Q = C \cdot i \cdot A / 360$
- Manning's trapezoidal sizing
- Per-segment cost → GeoPackage

Drainage design parameters (10-year return period)

- ▶ Manning's $n = 0.025$ (earthen) · side slope 1.5H:1V · rainfall $i = 50$ mm/hr · runoff $C = 0.65$
- ▶ Velocity window 0.3–2.0 m/s (self-cleaning, non-erosive) · cost rate depth-dependent per metre
- ▶ Every channel sized to carry its catchment's design discharge — zero capacity exceedances

How We Validate — Honest, Defensible Metrics

No fabricated accuracy: every number comes from a withheld or independent check

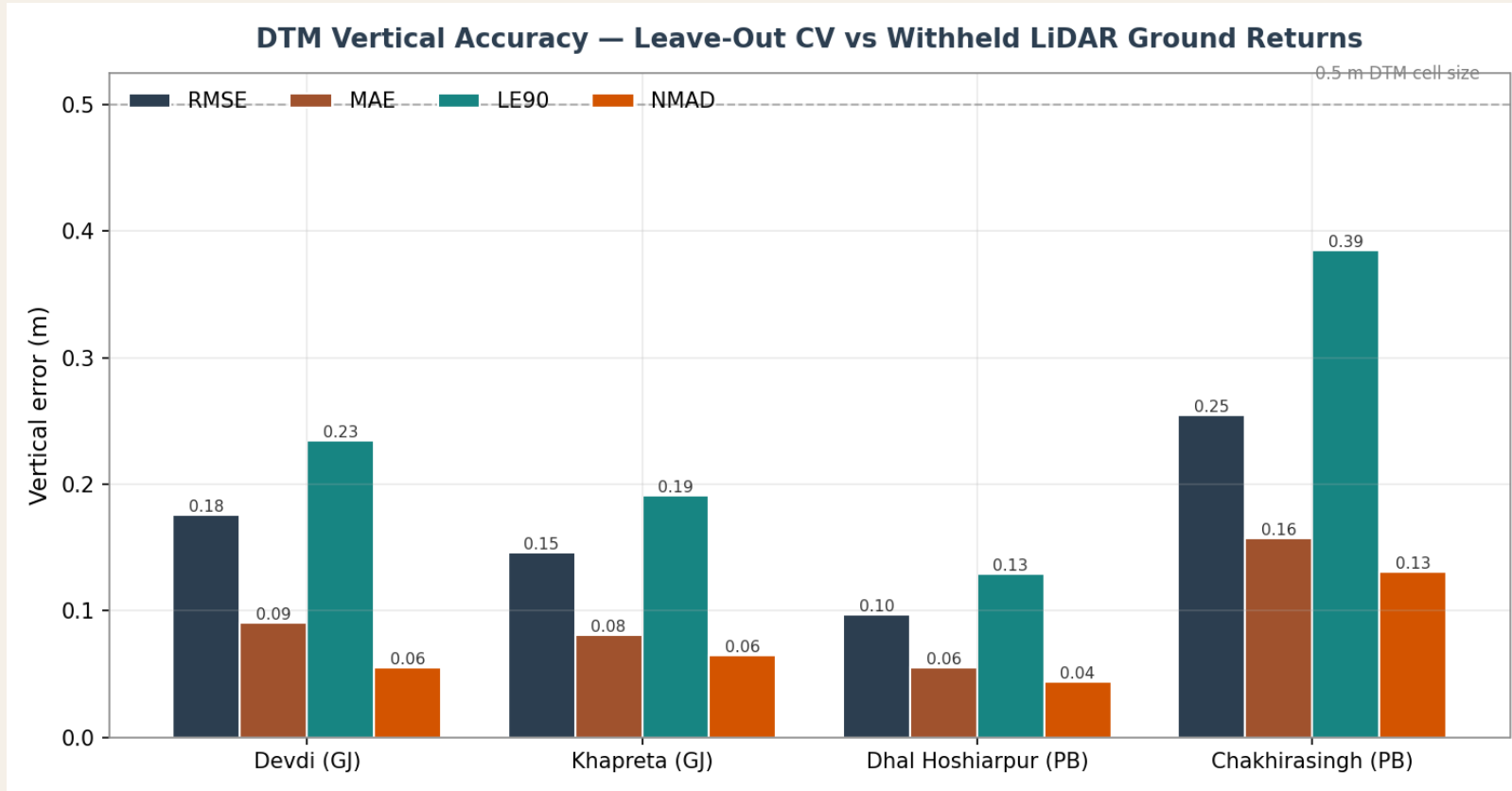
Output	Validation method	Why it is defensible
DTM accuracy	Leave-out cross-validation: interpolate withheld LiDAR ground returns and compare	True vertical error vs real measurements — ASPRS-style, not a self-comparison
Waterlogging	5-fold CV fidelity to a physically-derived risk index + cross-village transfer	Tests genuine generalisation across terrain, not memorised thresholds
Drainage	Hydraulic capacity check (Manning's) on every designed channel	Engineering constraint satisfaction, fully reproducible

Transparent about limits

- No observed flood records exist for these villages, so waterlogging labels are physically-derived, not ground-truthed — we report this openly.
- DTM has no external survey reference; we validate internally against the LiDAR returns themselves (the densest available truth).

Results — DTM Vertical Accuracy

Leave-out cross-validation against withheld LiDAR ground returns

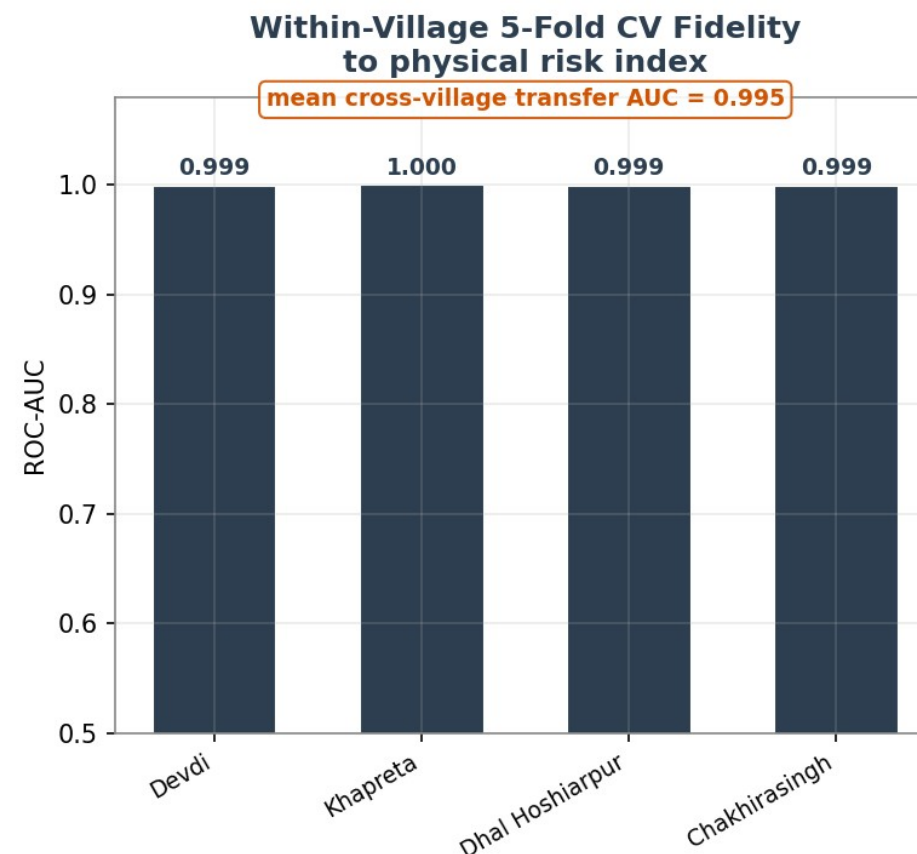
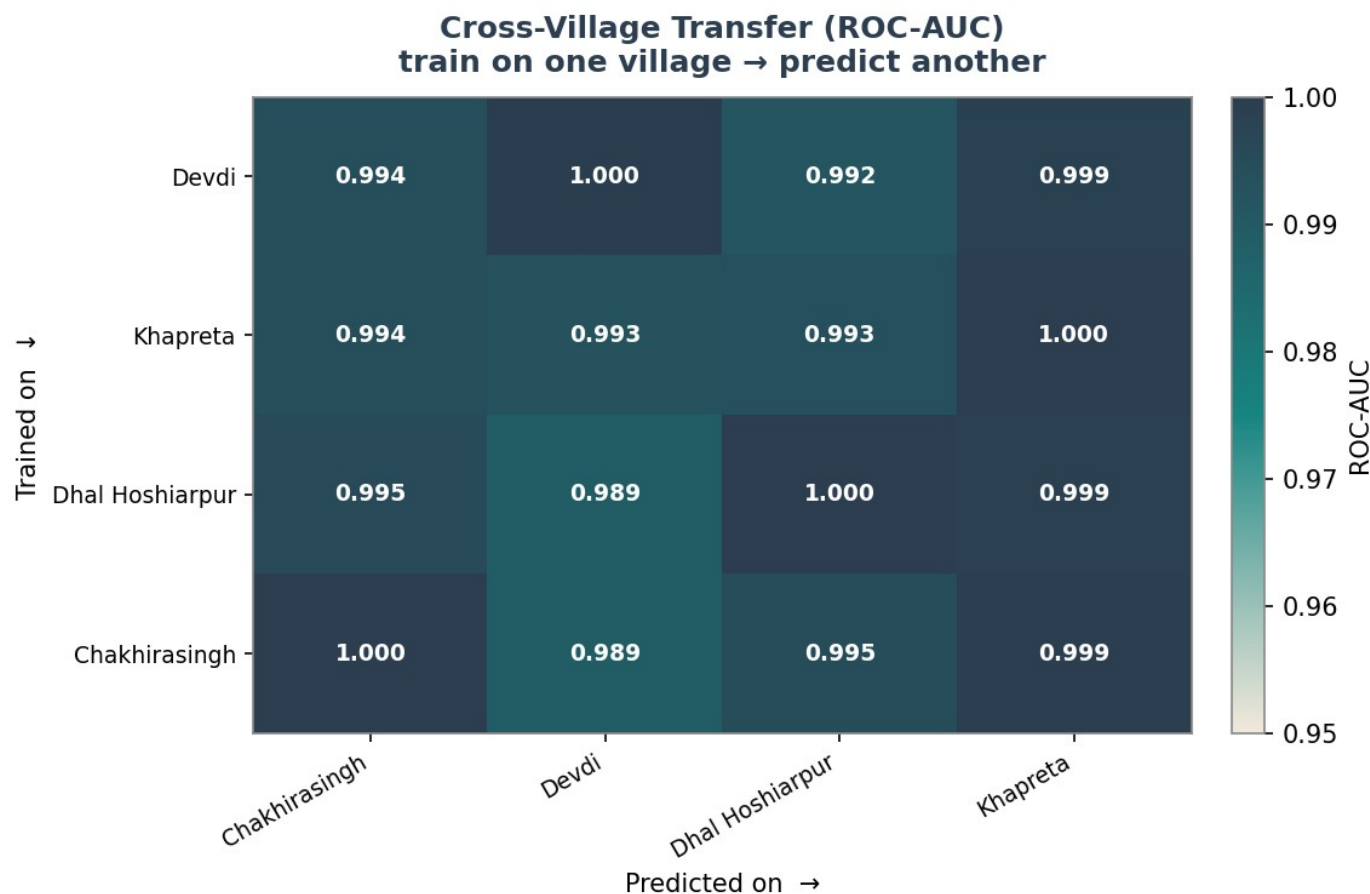


Key takeaways

- RMSE 0.10–0.25 m across villages
- LE90 (ASPRS) as low as 0.13 m
- Decimetre-level fit on dense terrain
- Errors track relief: flatter abadi areas are most accurate
- Same IDW method validated identically on every village

Results — Waterlogging Risk & Cross-Village Transfer

The honest generalisation test: train on one village, predict another

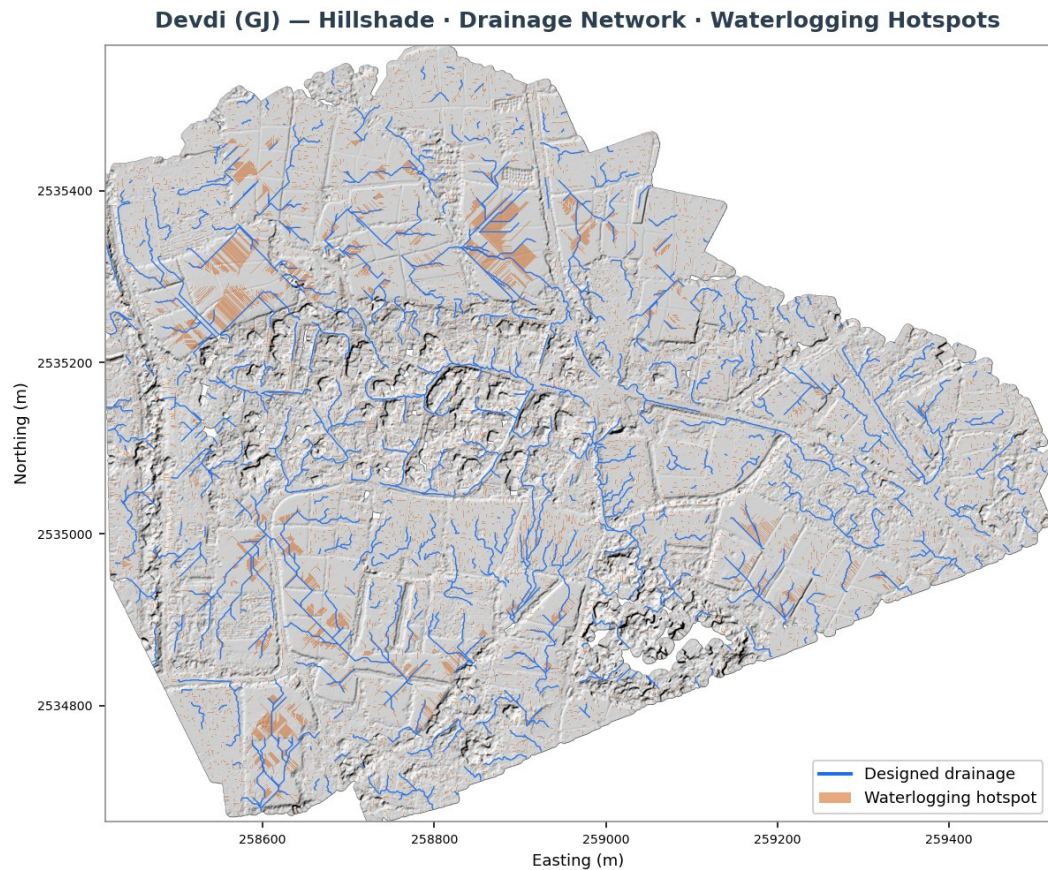


Strong off-diagonal AUC shows the model learns transferable terrain→risk structure, not village-specific thresholds. Next step: calibrate against observed flood events.

Per-Village Results — Drainage & Hotspots

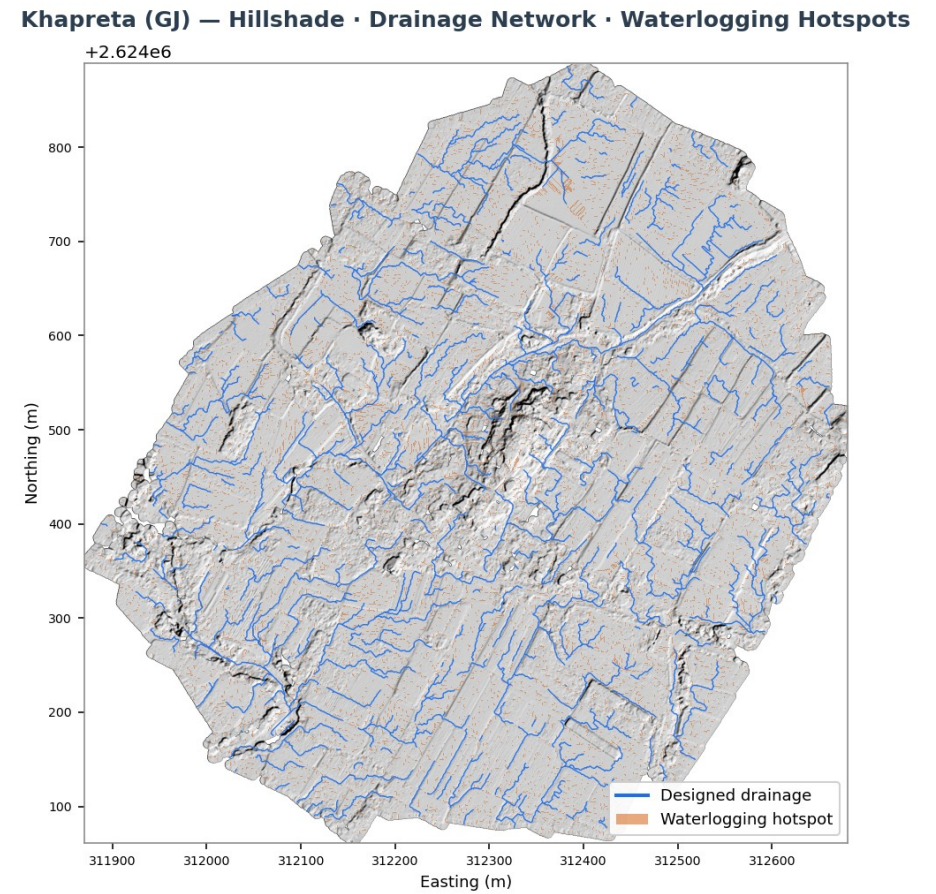
Devdi · Gujarat

DTM RMSE 0.1762 m · LE90 0.2349 m · WL AUC 0.9993
966 channels · 54.1 km · ₹434 L



Khapreta · Gujarat

DTM RMSE 0.1458 m · LE90 0.1914 m · WL AUC 0.9999
575 channels · 27.8 km · ₹233 L

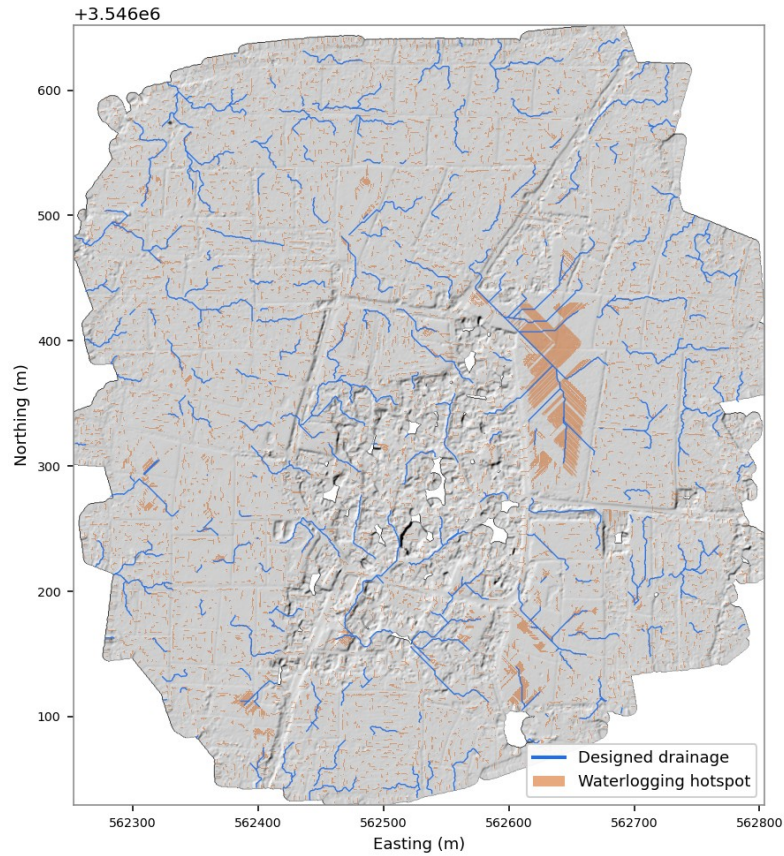


Per-Village Results — Drainage & Hotspots

Dhal Hoshiarpur · Punjab

DTM RMSE 0.0969 m · LE90 0.1289 m · WL AUC 0.9994
297 channels · 9.8 km · ₹79 L

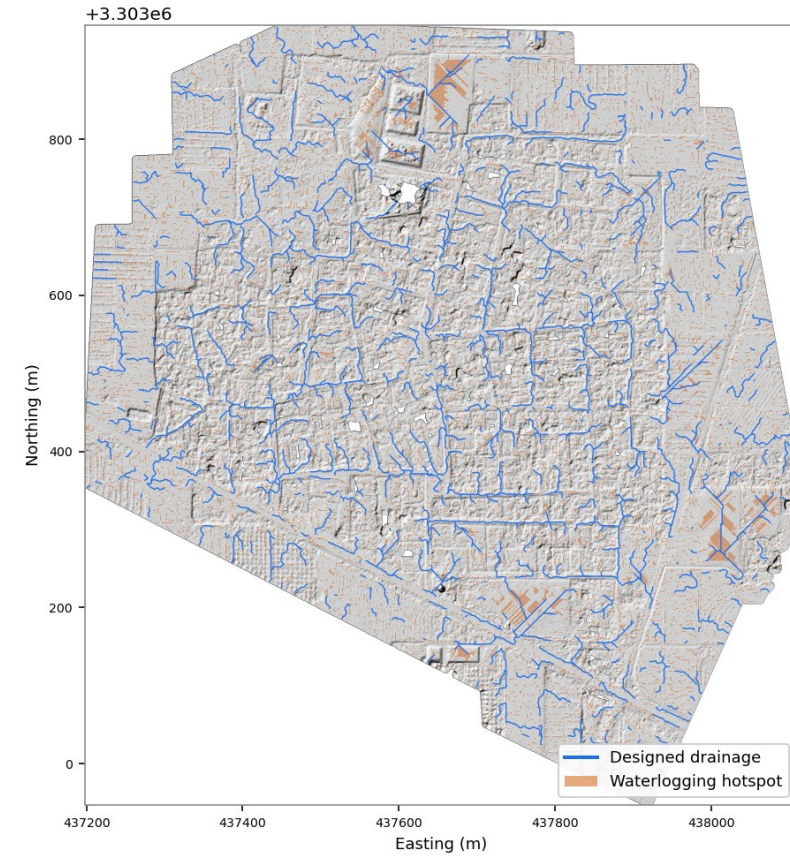
Dhal Hoshiarpur (PB) — Hillshade · Drainage Network · Waterlogging Hotspots



Chakhirasingh · Punjab

DTM RMSE 0.2544 m · LE90 0.3851 m · WL AUC 0.9989
1013 channels · 43.7 km · ₹351 L

Chakhirasingh (PB) — Hillshade · Drainage Network · Waterlogging Hotspots



Results Scorecard — All Villages

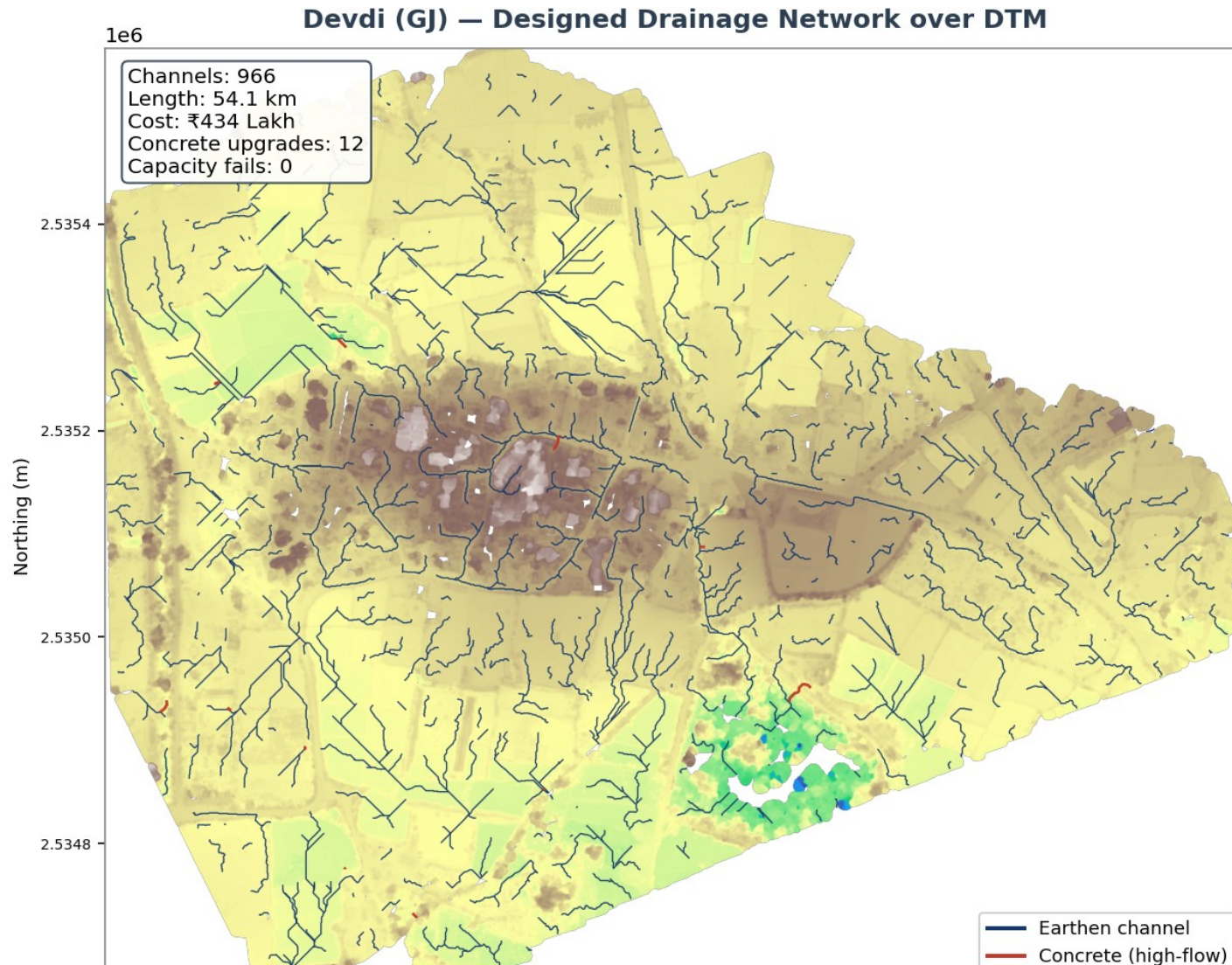
Per-Village Results Scorecard — Honest Metrics

Village	Ground pts	Relief (m)	DTM RMSE (m)	DTM LE90 (m)	WL AUC	Channels	Drain (km)	Cost (₹L)
Devdi	49.6M	23.3	0.18	0.23	0.999	966	54.1	434
Khapreta	148.3M	35.8	0.15	0.19	1.000	575	27.8	233
Dhal Hoshiarpur	28.7M	8.5	0.10	0.13	0.999	297	9.8	79
Chakhirasingh	5.6M	18.9	0.25	0.39	0.999	1013	43.7	351

One pipeline, one config — identical processing and validation applied to every village across two states.

Drainage Network Design — Worked Example

Devdi: MST-routed, hydraulically-sized, fully costed



Design summary

Metric	Value
Channels	966
Total length	54.1 km
Total cost	₹434 Lakh
Avg velocity	0.543 m/s
Capacity fails	0

Channels follow natural flow paths (minimum spanning tree), each sized by Manning's equation to carry its catchment's 10-yr design storm.

Scalability & Production Readiness

Handles any size

- Auto-tiling with overlap buffer
- Chunked streaming I/O
- Tested up to 170 M+ points
- Memory-safe on commodity hardware

Transfers across regions

- Validated in Gujarat & Punjab
- Cross-village model transfer shown
- One config, many villages
- No per-site manual tuning

Open & integrable

- Cloud-Optimized GeoTIFF rasters
- GeoPackage vector layers
- Loads directly in QGIS/ArcGIS
- OGC-compliant for govt GIS

Limitations & Roadmap

What we would do with more data and time — stated plainly

Current limitations

- Waterlogging labels are physically-derived, not yet validated against observed floods.
- DTM lacks an independent DGPS survey reference; validated against the LiDAR returns.
- Ground-refinement gains over SMRF are marginal on flat abadi terrain.
- 4 villages processed; pipeline is ready to run on the remaining SVAMITVA tiles.

Roadmap

- Calibrate risk model on real flood-event / drain-complaint records (expected AUC $\uparrow$).
- DGPS checkpoints for absolute DTM accuracy (target $LE90 \leq 0.15$ m).
- Roll the batch pipeline across all 10 villages and beyond.
- Add rainfall-runoff hydrograph routing for peak-flow design.
- Web dashboard for planners to inspect and edit networks.

Thank You

DTM Drainage AI — MoPR Geospatial Intelligence Hackathon

charansaiponnada · IIT Tirupati

Automated DTM · Waterlogging Risk · Drainage Design from LiDAR point clouds

Built end-to-end · Validated honestly · Ready to scale